ALAGAPPA UNIVERSITY

(Reaccredited by NAAC with 'A' Grade, CGPA - 3.21/4)

KARAIKUDI-630 003, TAMIL NADU, INDIA

DEPARTMENT OF BIOINFORMATICS

Organizing

4th National Symposium cum Workshop on

"Recent Trends in Structural Bioinformatics and Computer Aided Drug Design"

[SBCADD'2012]

February 20th - 23rd, 2012

Venue

L. CT. L. Palaniappa Chettiar Memorial Auditorium, Alagappa University

Sponsored by

Alagappa University

&

Department of Biotechnology (DBT)

Government of India

Please log on to SBCADD'2012 for further information

- **Dr. S. Parthasarthy**
  Bharthidasan University, Trichy
- **Dr. Ritushree Kukreti**
  IGIB, Delhi
- **Dr. K. Sekar**
  IISc, Bangalore
- **Dr. V. Subramanian**
  CLRI, Chennai
- **Dr. N. Sundara Baalaji**
  Bharathiar University, Coimbatore
- **Dr. Sunil Kumar**
  Institute of Life Sciences, Bhubaneswar
- **Dr. M. Suresh Kumar**
  Pondicherry University, Puducherry
- **Dr. K. Thangaraj**
  CCMB, Hyderabad
- **Dr. D. Velmurugan**
  University of Madras, Chennai
- **Mr. R. Raghu**
  Schrödinger GmbH, USA

PROGRAMME COMMITTEE

PATRON

Prof. S. Sudalaimuthu
Vice-Chancellor

CO-PATRONS

Dr. K. Manimekalai
Registrar i/c

Prof. S. Karutha Pandian
Chairperson, School of Biological Sciences

CONVENER

Prof. J. Jeyakanthan
Professor and Head, Department of Bioinformatics

ORGANIZING SECRATARY

Dr. M. Karthikeyan
Assistant Professor, Department of Bioinformatics

ORGANIZIN G COMMITTEE MEMBERS

Dr. Sanjeev Kumar Singh
Reader, Department of Bioinformatics

Dr. P. Srinivasan
Assistant Professor, Department of Bioinformatics

REGISTRATION

The participants are requested to send the filled in registration form along with registration fee on or before **10th February, 2012**. The DD should be drawn in favour of Organizing Secretary, payable at Karaikudi.

Students	: Rs.	500
Research Scholars	: Rs.	750
Faculties from Academic Institution	: Rs.	1000
Faculties from Non Academic Institution / Industries	: Rs.	1500

ACCOMMODATION

Request for accommodation should be sent to the organizing secretary at the earliest opportunity to the students in the University Hostel/Hotel. The expenses for accommodation and food will be met by the participants.

HOTELS AVAILABLE AT KARAIKUDI

Subalakshmi	+91 4565 235200/01/02/03
L.T. Lodge	+91 4565 238792
M.R. Lodge	+91 4565 233796
Jagan	+91 4565 227273,225446
SMS lodge	+91 4565 227741,230206
Nachiapan	+91 4565 230077

HOW TO REACH ALAGAPPA UNIVERSITY

Train and bus facilities are frequently available in Karaikudi. For further details please log on to www.mustseeindia.com/Karaikudi/how-to-reach

ADDRESS FOR COMMUNICATION

Dr. M. Karthikeyan
Organizing Secretary, SBCADD' 2012
Department of Bioinformatics
Science Block, 4th Floor
Alagappa University, Karaikudi - 630 004
Tamil Nadu, India.
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ALAGAPPA UNIVERSITY

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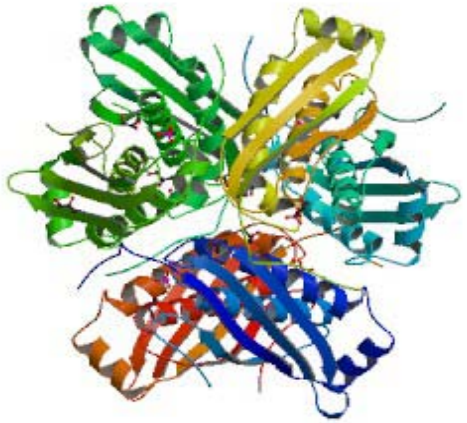
DEPARTMENT OF BIOINFORMATICS

4th National Symposium cum Workshop on
"Recent Trends in Structural Bioinformatics and Computer Aided Drug Design" [SBCADD'2012]

20th - 23rd February, 2012



Venue

L.CT.L. Palaniappa Chettiar Memorial Auditorium

Sponsors

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Government of India

THEME OF THE SYMPOSIUM AND WORKSHOP

Human Genome Project is largely complete, so the problem facing by scientists is how to maximize the use of the newly acquired data for improving health care. It is estimated that the number of therapeutic targets available for drug discovery will increase from the current number of 600 to 1000 to perhaps as many as 5000 to 10,000. In addition to the challenges that this number provides to the pharmaceutical industry, there is the issue of sequence variation through Single Nucleotide Polymorphisms (SNPs), some of which may impact upon the way that the body handles drug treatment. This may be due to a direct effect on the binding site of the protein target through non-conservative alterations of the amino acid sequence, or else through indirect effects on drug metabolizing enzymes. In order to meet these challenges, the combination of structural proteomics and computer-aided small molecule design provides opportunities for creating new molecules *in silico*; these may be designed to bind to selected pharmacogenetic variants of a protein in order to overcome the non-responsiveness of certain patient groups to a particular medicine.

In silico drug discovery methods reduce both time and cost of development of novel drug/lead molecules. Knowledge about the molecule, its interaction with the drug, molecular modeling, and new drug development creates an awareness of the molecule at the organ level and aids in the prevention, diagnosis, and treatment of diseases. Computational tools have the advantage of delivering new lead candidates more quickly and at lower cost. In the 21st century, *in silico* methods facilitate target identification, structure prediction and lead/drug discovery. The computational methods expect more reliable and expeditious protocols for development of novel ideas, which increase the potential leads. The computer-assisted molecular design has succeeded in the QSAR and protein modeling algorithms to improve activity of lead compounds. Selecting candidates *via* the described *in silico* virtual screening, should help to reduce the list of candidate molecules and thereby reduce time and costs significantly.

ABOUT OUR UNIVERSITY

Alagappa University was founded by the great Philanthropist and Educationist Padma Bhushan Dr. RM. Alagappa Chettiar in 1985 at Karaikudi, Tamilnadu, India. Alagappa University is recognized by the UGC, Government of India and was reaccredited with "A" grade by NAAC. Wide shredded 420 acres green campus with all potential academic activities. The post graduate and research departments in Science, Arts, Humanities and other disciplines in the Alagappa University offers more novel and innovative programmes through regular, distance participation and e-learning. Alagappa University educates nearly 0.1 million students/people every year especially in the rural and backward areas of Tamil Nadu. The motto of Alagappa University is to develop leadership in various disciplines through **EXCELLENCE IN ACTION**.

ABOUT THE DEPARTMENT

Department of Bioinformatics was started in the year 2008, to facilitate the teaching and research in the interdisciplinary areas of Structural Bioinformatics, Computational Genomics and Proteomics to create an intellectual environment for interdisciplinary education and training. Our department comprises of well experienced faculty members with research interests and expertise in Small and Macro Molecule X-ray Crystallography, Structural Biology, Molecular Modelling, Computer Aided Drug Design, Bio-computing, Pharmacogenomics, Chemoinformatics, Molecular Cell Biology, Genetic Engineering, Genomics, Proteomics, Quantum Chemistry, Database and Software development. We apply our computational and experimental techniques to study the interaction of small chemical compounds with proteins and nucleic acids and to characterize their molecular mechanisms towards the novel discovery for human diseases. Our dynamic environment is focused on students to gain experience in these fields through academic, industrial rotations and internship programs. The Department of Bioinformatics is offering a two year M.Sc., degree course and one year M. Phil., course in Bioinformatics. Doctoral programme is also being offered in Bioinformatics.

SCIENTIFIC TOPICS

- Small and Macromolecule X-ray Crystallography
- Protein Modeling and Molecular Docking
- Molecular Mechanics & Dynamics
- Pharmacophore Modeling & 3D-QSAR
- Free Energy Calculation Methods
- *De Novo* Drug Design
- Quantum Chemistry
- Functional Genomics
- Pharmacogenomics
- Database Creation and Development
- Text and Data mining

PARTICIPANTS

Students, Research Scholars, Research Scientists and Faculty members from the Universities, Colleges, Institutes and Industries.

CALL FOR POSTER PRESENTATIONS

The prospective participants are requested to submit the extended abstracts on various aspects on the theme of the symposium (not exceeding 250 words) to the organizing secretary on or before **10th February, 2012**. The abstract must be sent to **bioinfoau@gmail.com**. Kindly visit the university website for further information.

LIST OF SPEAKERS

- **Dr. Arpita Yadav**
  Institute of Engineering and Technology, Kanpur
- **Dr. U. Deva Priyakumar**
  IIIT, Hyderabad
- **Dr. K. Gunasekaran**
  University of Madras, Chennai
- **Dr. G. Jayaraman**
  University of Madras, Chennai
- **Dr. R. Krishna**
  Pondicherry University, Puducherry
- **Dr. S. Krishnaswamy**
  Madurai Kamaraj University, Madurai
- **Dr. M. Michael Gromiha**
  IIT, Chennai

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REGISTRATION FORM

(To be filled in block letters)

Name:_____

Sex: _____ Designation:_____

Institutional Affiliation with postal Address: _____

_____

_____

Telephone: _____ Email: _____

Are you presenting poster *(Please click)*: Yes ◯ No ◯

Title of the Paper: _____

_____

_____

Whether accommodation requested *(Please click)*: Yes ◯ No ◯

Hotel ◯ University hostel ◯

Particulars of Fee:

Amount Rs._____ DD No: _____

Date: _____ Name of the Bank: _____

Signature of Applicant **Head of the Department/ Research Supervisor**

Note: Abstract has to be accompanied by duly filled in registration form along with DD. The DD should be drawn in favour of **Organizing Secretary**, payable at **Karaikudi**.